
Unit – 2 : Session – 12 : SLO – 1 | Mutex Locks — Pseudocode Completion

Answers

TODO 1: acquire(lock) (lock the mutex before entering the critical section).

TODO 2: release(lock) (unlock the mutex after leaving the critical section).

TODO 3: join(thread1); join(thread2) (wait for both threads to complete).

Result: each of the 2 threads adds 5, so the final counter value is 10 (no lost updates because the mutex gives mutual exclusion).

Completed pseudocode

```
shared int counter = 0
mutex lock

function increment_thread():
    for i from 1 to 5:
        acquire(lock)                // TODO 1
        // Critical Section
        temp = counter
        temp = temp + 1
        counter = temp
        print("Counter value: ", counter)
        release(lock)                // TODO 2

thread1 = create_thread(increment_thread)
thread2 = create_thread(increment_thread)
join(thread1)                        // TODO 3
join(thread2)
print("Final counter value: ", counter)
```